

Data Science Applications to Politics Research

Week 2 Seminar

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GV330

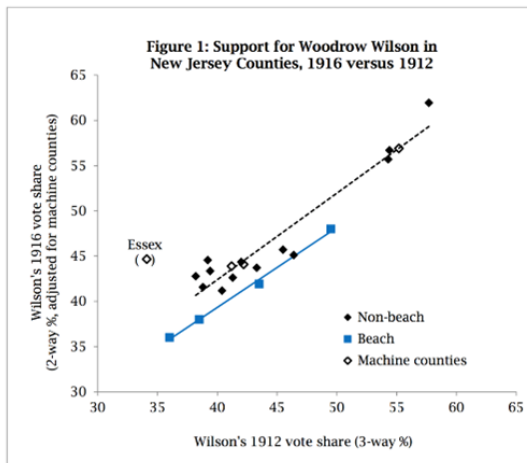
“If you torture the data long enough, it will confess.” - Ronald Coase

- **Opportunity:** Researchers have many “degrees of freedom” in the design and analysis of a study → p-hacking (may not always be intentional, see Gelman & Loken 2013)
- **Motive:** Researchers have incentives (from journals, tenure requirements, etc.) to find statistical significance
- **Result:** Biased evidence base (also contributes to replication crisis)

Shark attacks and voting (Achen & Bartels 2017)



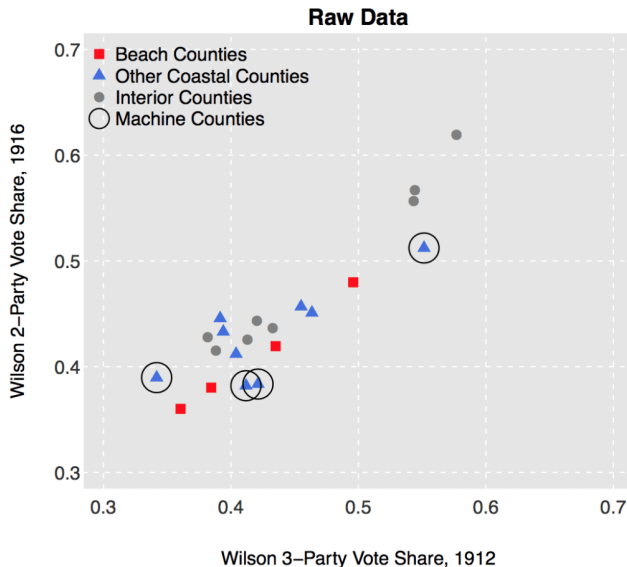
Shark attacks and voting (Achen & Bartels 2017)



“A dramatic series of shark attacks in NJ in 1916... voters significantly punished the incumbent president at the polls”

Achen, C., & Bartels, L. (2017). *Democracy for Realists*. Princeton University Press.

Fowler & Hall 2018: Re-analysis of Achen & Bartels



Critique of Achen & Bartels (Fowler & Hall 2018)

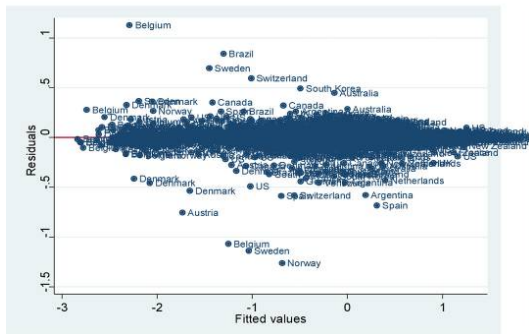
- **Correlation**: election results from different counties are not independent events (can inflate our sense of the “true” sample size)
- **‘Forking paths’**: many different ways to analyze the data (e.g., choice of how to treat “outliers”)

“We assemble data on **every fatal shark attack in U.S. history** and county-level returns from every presidential election between 1872 and 2012, and we find **little systematic evidence that shark attacks hurt incumbent presidents or their party.**”

Fowler, A., & Hall, A. B. (2018). Do shark attacks influence presidential elections? Reassessing a prominent finding on voter competence. *The Journal of Politics*, 80(4), 1423-1437.

Critique of Shin 2019 (Choi 2022)

Model 1 (Shin's Model 1)



- Takeaway: OLS slope + p -value can be **highly sensitive** to one influential case.
- Choi (2022) argues Norway drives the “oil wealth → restrictive immigration” finding in a published claim.
- Lesson for us: always run **influence diagnostics** (leverage/Cook's D) and test robustness (e.g., robust regression).

Choi SW. Outlier analysis: Natural resources and immigration policy. *PLoS One*. 2022 Jan 13;17(1):e0261533.

Go to
<http://shinyapps.org/apps/p-hacker/>

Schönbrodt, F. D. (2016). p-hacker: Train your p-hacking skills!

Set up your study

p-hacker: Train your p-hacking skills!

Begin at the "New Study" tab



Manual

New study

[Now: p-hack!](#)

No study run yet - click on 'Run new experiment' at the bottom of the left panel!

Settings for initial data collection:

Name for experimental group

Elderly priming

Name for control group

Control priming

Initial # of participants in each group



True effect in population



Number of DVs



Run new experiment

(Discards previous data)

Decide how many participants you want to collect initially.

Pro-Tip: You increase your chances of finding a significant effect when you run many studies with many participants, instead of few studies with many participants (Bakker, van Dijk, & Wicherts, 2012)!

Next, decide what the true effect size should be.

Pro-Tip: For a proper training in p-hacking, always select "0"! Then you can train to squeeze out an effect from nothing - isn't that cool!?

Next, decide how many potential dependent variables (DVs) you assess. (Technical detail: all DVs correlate to $r=.5$)

Pro-Tip: The more DVs you measure, the more you increase the chance of finding something! DV_all is an aggregate of all DVs.

Finally: Run your experiment!

Inspect results of your study

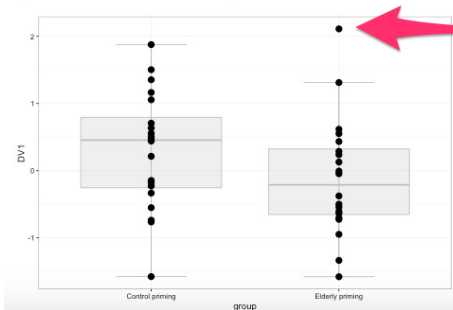
Tests for each DV

Name	N	Statistic	p-Value	Sign.	Actions
DV1	40	$F(1, 38) = 2.26$	$p = .141$		Save
DV2	40	$F(1, 38) = 0.38$	$p = .543$	ns	Save
DV3	40	$F(1, 38) = 0.03$	$p = .859$	ns	Save
DV4	40	$F(1, 38) = 0.73$	$p = .400$	ns	Save
DV_all	40	$F(1, 38) = 0.1$	$p = .750$	ns	Save

Not too bad! $p = .14$ in your first DV is a clear trend towards significance. (Discard the other DVs, focus on DV1)

Choose DV to plot

DV1 ▼



Here's a clear outlier (you can justify its exclusion with the box plot rule).

Try to click on a data point to exclude it!

Can't get a significant result? "Now p-hack!"

New study

Now: p-hack!

Basic tools to improve your p-value:

- ☒ Control for age
- ☒ Control for gender
- ☐ Interaction with gender

Add 5 new participants

Add 10 new participants

Tests for each DV

Name	N	Statistic	p-Value	Sign.	Actions
DV1	70	$F(1, 66) = 0.67$	$p = .417$	ns	<button>Save</button>
DV2	70	$F(1, 66) = 1.98$	$p = .164$	ns	<button>Save</button>
DV3	70	$F(1, 66) = 1.39$	$p = .243$	ns	<button>Save</button>
DV4	70	$F(1, 66) = 0.66$	$p = .419$	ns	<button>Save</button>
DV_all	68	$F(1, 64) = 4.77$	$p = .033$	*	<button>Save</button>

Choose DV to plot

DV_all

A similar tool

See also <https://projects.fivethirtyeight.com/p-hacking/>

You're a social scientist with a hunch: **The U.S. economy is affected by whether Republicans or Democrats are in office.** Try to show that a connection exists, using real data going back to 1948. For your results to be publishable in an academic journal, you'll need to prove that they are "statistically significant" by achieving a low enough p-value.

1 CHOOSE A POLITICAL PARTY

☒ Republicans ☐ Democrats

2 DEFINE TERMS

Which politicians do you want to include?

☒ Presidents
☐ Governors
☐ Senators
☐ Representatives

How do you want to measure economic performance?

☒ Employment
☐ Inflation
☐ GDP
☒ Stock prices

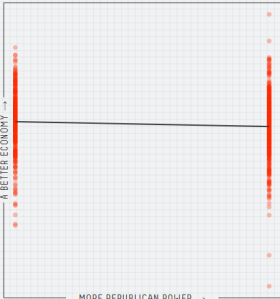
Other options

☒ Factor in power
Weight more powerful positions more heavily

☐ Exclude recessions
Don't include economic recessions

3 IS THERE A RELATIONSHIP?

Given how you've defined your terms, does the economy do better, worse or about the same when more Republicans are in power? Each dot below represents one month of data.



4 IS YOUR RESULT SIGNIFICANT?

If there were no connection between the economy and politics, what is the probability that you'd get results at least as strong as yours? That probability is your p-value, and by convention, you need a p-value of 0.05 or less to get published.



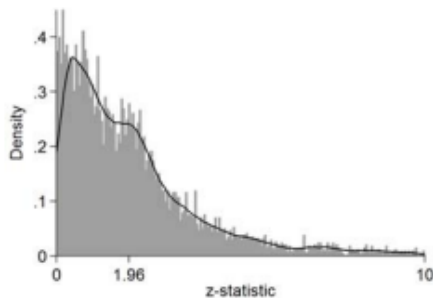
Result: Publishable

You achieved a p-value of **0.03** and showed that **Republicans** have a **negative effect** on the economy. Get ready to be published!

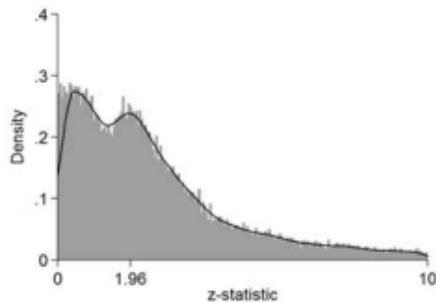
If you're interested in reading real (and more rigorous) studies on the connection between politics and the economy, see the work of Larry Bartels and Alan Blinder and Mark Watson.

Data from The @unitedstates Project, National Governors Association, Bureau of Labor Statistics, Federal Reserve Bank of St. Louis and Yahoo Finance.

Evidence of P-Hacking:



(a) Lab. experiments or RCT data.



(b) Other data.

Identify 34 key researcher degrees of freedom (see article for full list):

Table 1

Checklist for different types of degrees of freedom in the planning, executing, analyzing, and reporting of psychological studies

Code	Related	Type of Degrees of Freedom
Hypothesizing		
T1	R6	Conducting explorative research without any hypothesis
T2		Studying a vague hypothesis that fails to specify the direction of the effect
Design		
D1	A8	Creating multiple manipulated independent variables and conditions
D2	A10	Measuring additional variables that can later be selected as covariates, independent variables, mediators, or moderators
D3	A5	Measuring the same dependent variable in several alternative ways
D4	A7	Measuring additional constructs that could potentially act as primary outcomes
D5	A12	Measuring additional variables that enable later exclusion of participants from the analyses (e.g., awareness or manipulation checks)
D6	C4	Failing to conduct a well-founded power analysis
D7		Failing to specify the sampling plan and allowing for running (multiple) small studies

...

Misconduct & Fraud: Rare(?) but serious

- **Includes:** Falsifying some or all data and/or results, as well as plagiarism and other forms of misconduct
- **Result:** False or biased evidence base
- **Note:** Fabrication of data (e.g., LaCour & Green 2014) less common than other “questionable research practices”

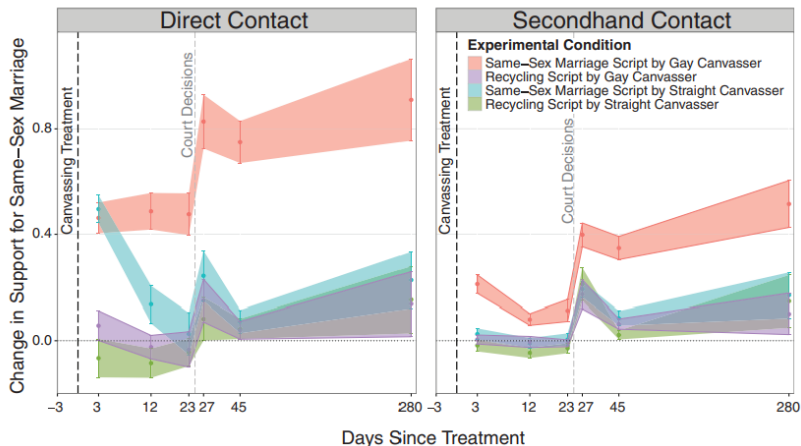


Fig. 1. Direct and secondhand effects on support for same-sex marriage, by assigned message and messenger, and time since treatment. The first vertical dashed line represents the canvassing intervention, which was administered between Internet survey waves 1 and 2. The second vertical dashed line represents the U.S. Supreme Court decisions striking down California's ban on same-sex marriage. The y axis is opinion change between the baseline survey and subsequent survey waves, with higher scores indicating more support for same-sex marriage. Points represent mean values, and bars display

Fraud in Dishonesty Studies (Shu, Mazar, Gino, Ariely, & Bazerman (2012))

SIGN AT THE BOTTOM

Shu et al. 10.1073/pnas.1209746109

Form 3305		Research Study Tax Return		Keep a copy of this return for your records.
(Rev. June 2010)		For the period June 1, 2010, through August 30, 2010		OMB No. 1545-0011
Center for Decision Research				
Write Clearly		Name		PID
		Address (Number, street, and room or suite number)		For Administrative Use Only
		City, State, and ZIP code		T RP IP I TL
Part 1 Please fill out the questions below to compute your taxed payment.				
1. Please enter the payment you received on the problem solving task (31 per current status you solved in the other round).....				1
2. Tax on payment: Please enter the equivalent of a 20% tax on your payment (i.e., 20 cents for every dollar earned).....				2
3. Please subtract the value specified in box 2 from value specified in box 1.....				3
Part 2 Participants will be compensated for extra expenses they have incurred in order to participate in this study. In Part 2, you are asked to estimate the costs incurred in order to participate. These costs will be deducted from your tax return.				
1. Please estimate the time it took you to come to the lab. You will be compensated \$0.10 per minute, up to a 2 hour maximum.....				4
2. Please estimate the cost of your commute, if any, to come to the lab. You will be compensated up to a maximum of \$0.2.....				5
3. Please add the value specified in box 4 and the value specified in box 5.....				6
Part 3 Please compute your final payment.				
1. Please add the value specified in box 3 and the value specified in box 6. This is the amount of your final payment for today's session.....				7
Sign Here				
I declare that I carefully examined this return and that to the best of my knowledge and belief it is correct and complete.				
Signature _____ Date _____				

Fig. S5. Tax form used in experiment 1, signed at the bottom condition.

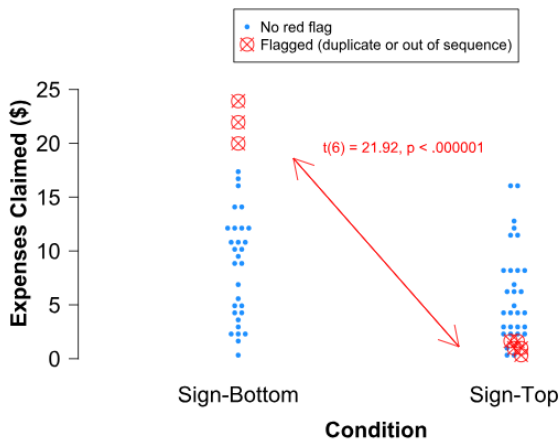
SIGN AT THE TOP

Form 3305		Research Study Tax Return		Keep a copy of this return for your records.
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2. Please estimate the cost of your commute, if any, to come to the lab. You will be compensated up to a maximum of \$0.2.....				5
3. Please add the value specified in box 4 and the value specified in box 5.....				6
Part 3 Please compute your final payment.				
1. Please add the value specified in box 3 and the value specified in box 6. This is the amount of your final payment for today's session.....				7
Sign Here				
I declare that I carefully examined this return and that to the best of my knowledge and belief it is correct and complete.				
Signature _____ Date _____				

Suspicious observations drive effect

Flagged Observations Show Huge Effect

Travel Expenses in Study 1 - Shu et al. (2012)



<https://datacolada.org/109>

COMMENT

Open Access



Gene name errors are widespread in the scientific literature

Mark Ziemann¹, Yotam Eren^{1,2} and Assam El-Osta^{1,3*}

Abstract

The spreadsheet software Microsoft Excel, when used with default settings, is known to convert gene names to dates and floating-point numbers. A programmatic scan of leading genomics journals reveals that approximately one-fifth of papers with supplementary Excel gene lists contain erroneous gene name conversions.

Keywords: Microsoft Excel, Gene symbol, Supplementary data

Abbreviations: GEO, Gene Expression Omnibus; JIF, journal impact factor

The problem of Excel software (Microsoft Corp., Redmond, WA, USA) inadvertently converting gene symbols to dates and floating-point numbers was originally described in 2004 [1]. For example, gene symbols such as

frequently reused. Our aim here is to raise awareness of the problem.

We downloaded and screened supplementary files from 18 journals published between 2005 and 2015 using a suite of shell scripts. Excel files (.xls and .xlsx suffixes) were converted to tabular separated files (tsv) with *ssconvert* (v1.12.9). Each sheet within the Excel file was converted to a separate tsv file. Each column of data in the tsv file was screened for the presence of gene symbols. If the first 20 rows of a column contained five or more gene symbols, then it was suspected to be a list of gene symbols, and then a regular expression (regex) search of the entire column was applied to identify gene symbol errors. Official gene symbols from Ensembl version 82, accessed November 2015, were obtained for *Arabidopsis thaliana*, *Caenorhabditis elegans*, *Drosophila melanogaster*, *Danio rerio*, *Escherichia coli*, *Gallus gallus*, *Homo sapiens*, *Mus musculus*, *Oryza sativa* and *Saccharomyces cerevisiae* [2]. The regex search used was

Is extrasensory perception (ESP) real?



Feeling the Future: Experimental Evidence for Anomalous Retroactive Influences on Cognition and Affect

Daryl J. Bem
Cornell University

The term *psi* denotes anomalous processes of information or energy transfer that are currently unexplained in terms of known physical or biological mechanisms. Two variants of *psi* are *precognition* (conscious cognitive awareness) and *premonition* (affective apprehension) of a future event that could not otherwise be anticipated through any known inferential process. Precognition and premonition are themselves special cases of a more general phenomenon: the anomalous retroactive influence of some future event on an individual's current responses, whether those responses are conscious or nonconscious, cognitive or affective. This article reports 9 experiments, involving more than 1,000 participants, that test for retroactive influence by "time-reversing" well-established psychological effects so that the individual's responses are obtained before the putatively causal stimulus events occur. Data are presented for 4 time-reversed effects: precognitive approach to erotic stimuli and precognitive avoidance of negative stimuli; retroactive priming; retroactive habituation; and retroactive facilitation of recall. The mean effect size (*d*) in *psi* performance across all 9 experiments was 0.22, and all but one of the experiments yielded statistically significant results. The individual-difference variable of stimulus seeking, a component of extraversion, was significantly correlated with *psi* performance in 5 of the experiments, with participants who scored above the midpoint on a scale of stimulus seeking achieving a mean effect size of 0.43. Skepticism about *psi*, issues of replication, and theories of *psi* are also discussed.

Fraud?

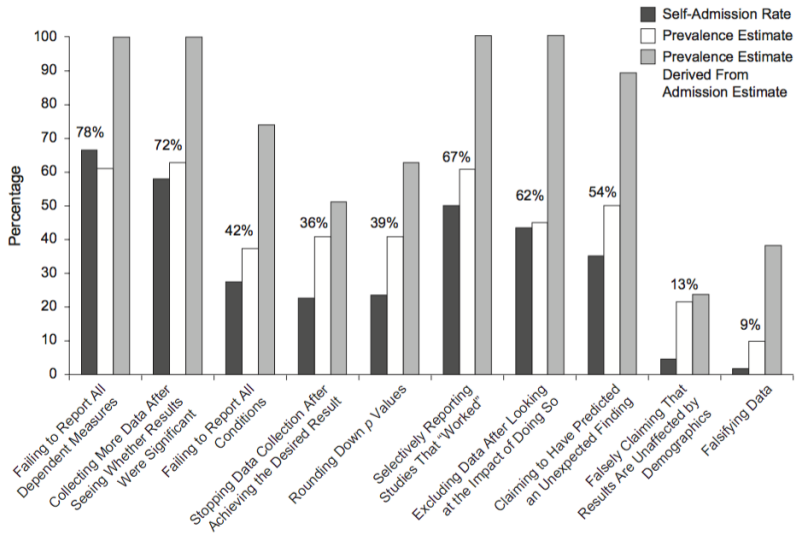
“I would start one [experiment], and if it just wasn’t going anywhere, I would abandon it and restart it with changes. . . I didn’t keep very close track of which ones I had discarded and which ones I hadn’t. . . .I was probably very sloppy at the beginning. I think probably some of the criticism could well be valid. I was never dishonest, but on the other hand, the critics were correct.”

—Daryl Bem

“I’m all for rigor, but I prefer other people do it. I see its importance—it’s fun for some people—but I don’t have the patience for it.”

—Daryl Bem, quoted in Slate Magazine

Survey of 2000 psychologists on questionable practices:



What about solutions?